

PharmaTwin

Formula Reference Guide

A plain-language guide to every calculation the platform runs — what it means, the formula itself, what each variable is, and a real worked example. Written for both plant/engineering staff and management.

How to use this guide: each formula is shown the same way — what it means in one sentence, the formula, what each variable stands for, and (where useful) a real example. Examples marked **Real batch example** use actual recorded data from **Batch PAR-084**, a genuine historical batch in the dataset — every number in those boxes is real, not invented.

Scope: Hyderabad Plant · Paracetamol 500mg. This guide covers deterministic calculations only — the two trained forecasting models' internal machine-learning math isn't reproduced here, only the formulas around them.

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01 Golden Batch & Golden Envelope

Everything else in this guide compares against one thing: the **Golden Batch**, a real, deliberately ideal batch. "Golden(t)" always means the Golden Batch's own actual recorded value at that same elapsed minute — never a flat, one-size-fits-all number.

Dynamic Golden Band

WHAT IT MEANS How far a live reading is allowed to drift from the Golden Batch at that exact minute before it gets flagged.

FORMULA

$$\begin{aligned}\text{Lower}(t) &= \text{Golden}(t) + \text{Lower Offset}(t) \\ \text{Upper}(t) &= \text{Golden}(t) + \text{Upper Offset}(t)\end{aligned}$$

VARIABLES

Golden(t)

The Golden Batch's real reading at minute t, in the parameter's own unit (°C, bar, L/min, etc.)

Lower Offset(t) ,
Upper Offset(t)

Pre-set margins around Golden(t) at minute t — wider during more variable phases, tighter during steady ones. The range is created from the Golden Batch value and the normal variation observed across 80 historical normal batches at each point in the process.

REAL BATCH EXAMPLE

At minute 200, the Golden Batch's real Temperature was **63.8°C**, with offsets of about -1.0°C / $+1.0^{\circ}\text{C}$ at that minute.

Allowed band = $62.8^{\circ}\text{C} - 64.8^{\circ}\text{C}$.

02 Quality & Stability

These two numbers feed almost everything downstream — Yield, Energy, and OEE all start from how *stable* the process was.

Parameter Status — Normal / Warning / Critical

WHAT IT MEANS How a live reading is labeled compared to its allowed band, from Section 01.

FORMULA

$\text{margin} = (\text{Upper} - \text{Lower}) \times 0.15$

```
if reading is outside [Lower, Upper]      → Critical
elif reading is within "margin" of an edge → Warning
else                                       → Normal
```

VARIABLES

Upper , **Lower** The band edges from Section 01, in the parameter's unit

0.15 The warning-zone width — 15% of the band, measured in from each edge

REAL BATCH EXAMPLE

Band = 62.8–64.8°C → width 2.0°C → margin = $0.15 \times 2.0 = 0.3^\circ\text{C}$. Warning zones: 62.8–63.1°C and 64.5–64.8°C.

Batch PAR-084's real Temperature reading at minute 200 was **64.7°C** — inside the upper Warning zone, so it was flagged **Warning**. This matches what the batch actually recorded.

Process Stability %

WHAT IT MEANS What share of the batch's running time Temperature, Pressure, and Flow Rate all stayed inside their normal operating range.

FORMULA

$\text{Stability \%} = \text{average of the 3 parameters'}$
 $\text{"\% of time spent in normal range"}$

VARIABLES

% in range Share of one-minute readings that fell between a parameter's spec lower and upper limit

REAL BATCH EXAMPLE

Batch PAR-084 — Temperature 72.0% in range, Pressure 100%, Flow Rate 100%.

Stability = $(72.0 + 100 + 100) \div 3 = 90.7\%$

Quality Score %

WHAT IT MEANS How close the batch's final drug-content result (assay) landed to the 100% target.

FORMULA

$$\text{Quality Score \%} = 100 \times (1 - \min(1, |\text{Assay\%} - 100| / 10))$$

VARIABLES

Assay% The batch's measured final assay result (target 100%, accepted spec range 95–105%)

10 The scoring width — twice the spec's half-width, so the score fades smoothly instead of hitting zero right at the spec edge

REAL BATCH EXAMPLE

Batch PAR-084's Assay = 99.37% $\rightarrow |99.37 - 100| = 0.63$

Quality Score = $100 \times (1 - 0.63/10) = 93.7\%$ (matches its real recorded 93.69%)

HOW ASSAY% ITSELF IS DERIVED (SAME FORMULA LIVE OR FINISHED)

"Dev" means: how far off that reading has run, on average, across the whole batch — not a single moment, and not the same live number shown on Process Monitoring. 0 = right on target, 1 = at the edge of normal.

$$\begin{aligned} \text{AssayRelevantDev} &= 0.6 \times \text{TempDev} + 0.2 \times \text{PressureDev} + 0.2 \times \text{FlowDev} + \\ &0.03 \times \text{FilterDev} + 0.02 \times \text{VibrationDev} + 0.02 \times \text{HumidityDev} \\ \text{Assay\%} &= 100 - (2.0 \times \text{AssayRelevantDev} \times 5.0) + \text{small noise} \end{aligned}$$

PAR-084's own scores: Temp 0.128, Pressure 0.047, Flow 0.053, Filter 0.13, Vibration 0.026, Humidity 0.023 $\rightarrow \text{AssayRelevantDev} \approx 0.098 \rightarrow \text{Assay\%} \approx 100 - 0.98 \approx 99.37\%$ (matches real value)

03 Yield

What share of the batch's theoretical maximum output was actually produced.

Yield %

WHAT IT MEANS Less stable processing means product is lost — Yield turns that loss into a percentage.

FORMULA

$$\begin{aligned} \text{Yield Loss} &= 0.35 \times (1 - \text{Stability}\% \div 100) \\ &\quad + 0.15 \times \text{AgitatorDev} + 0.03 \times \text{FilterPressureDev} + 0.02 \times \text{VibrationDev} \\ \text{Actual Output (kg)} &= \text{Theoretical Output (kg)} \times (1 - \text{Yield Loss}) \\ \text{Yield \%} &= 100 \times \text{Actual Output} \div \text{Theoretical Output} \end{aligned}$$

VARIABLES

Theoretical Output	150 kg — the fixed recipe amount for this product
Stability%	Process Stability % from Section 02
AgitatorDev, FilterPressureDev, VibrationDev	Same average drift meaning as Section 02

REAL BATCH EXAMPLE

Batch PAR-084 — Stability = 90.7%, Agitator drift = 0, Filter Pressure drift = 0.13, Vibration drift = 0.03

$$\text{Yield Loss} = 0.35 \times (1 - 0.907) + 0.15 \times 0 + 0.03 \times 0.13 + 0.02 \times 0.03 = 0.037$$

Actual Output = $150 \times (1 - 0.037) = 144.5 \text{ kg}$ (real recorded: 144.32 kg) → Yield = **96.21%** (matches exactly)

04 Energy & Efficiency

How much electricity a batch used in total, and how efficient that was per kilogram of product made.

Total Energy & Specific Energy Consumption (SEC)

WHAT IT MEANS A batch always burns a base amount of energy, plus extra when the process runs unstable.

FORMULA

$$\begin{aligned}\text{Instability Penalty} &= 0.5 \times (1 - \text{Stability}\% \div 100) \\ &\quad + 0.06 \times \text{FilterPressureDev} + 0.04 \times \text{HumidityDev} + \\ &\quad 0.02 \times \text{CompressedAirDev} \\ \text{Total Energy (kWh)} &= 180 + 0.16 \times \text{Duration (min)} \times (1 + \text{Instability Penalty}) \\ \text{SEC (kWh/kg)} &= \text{Total Energy} \div \text{Actual Output (kg)}\end{aligned}$$

VARIABLES

180 kWh	Fixed base energy draw for any batch, regardless of duration
0.16 kWh/min	Extra energy burned per minute the batch runs
Duration	Total batch run time, in minutes
FilterPressureDev, HumidityDev, CompressedAirDev	Same average drift meaning as Section 02

REAL BATCH EXAMPLE

Batch PAR-084 — Duration = 362 min, Stability = 90.7%, Filter Pressure drift = 0.13, Humidity drift = 0.02, Compressed Air drift = 0.03

$$\text{Instability Penalty} = 0.5 \times 0.093 + 0.06 \times 0.13 + 0.04 \times 0.02 + 0.02 \times 0.03 = 0.056$$

$$\text{Total Energy} = 180 + 0.16 \times 362 \times 1.056 = \mathbf{241.2 \text{ kWh}}$$
 (real recorded: 242.56 kWh)

$$\text{SEC} = 242.56 \div 144.32 = \mathbf{1.6808 \text{ kWh/kg}}$$
 (matches real recorded value)

WHAT THE INSTABILITY PENALTY MEANS

0 means the batch ran perfectly smoothly, adding no extra energy cost at all. The maximum possible penalty from instability alone is 0.5 (a 50% jump in energy use), before the smaller Filter Pressure/Humidity/Compressed Air terms add a bit more on top.

05 Batch / Production KPIs

The single headline number used on the Dashboard and Batch Explorer to summarize a batch's overall performance.

OEE % — Overall Equipment Effectiveness

WHAT IT MEANS One score combining how efficiently time was used, how well the equipment performed, and product quality.

FORMULA

$$\text{OEE \%} = (\text{Availability\%} \times \text{Performance\%} \times \text{Quality\%}) \div 10,000$$

VARIABLES

Availability% 100 × Golden Batch's duration ÷ this batch's duration, capped at 100% — a slower batch scores lower here

Performance% Not measured — a random placeholder within a severity-based range (90–98% Normal / 82–92% Warning / 75–88% Critical), since no real throughput sensor exists yet

Quality% The Quality Score % from Section 02

REAL BATCH EXAMPLE

Batch PAR-084 — Availability 100% (it finished faster than the Golden Batch's 383 minutes), Performance 88.4%, Quality 93.69%.

OEE = $(100 \times 88.4 \times 93.69) \div 10,000 = 82.8\%$ (matches real recorded 82.84%)

06 Prediction & Deviation Detection

While a batch is still running, these formulas turn a live trend into an early warning.

The two are easy to mix up, so each is tagged with the page it actually belongs to:

KPI Prediction Status and **Prediction Confidence** are about the whole batch's forecast final outcome (Yield, Quality, Energy, etc.) on the **KPI Prediction & Deviation Detection page**; **Time-to-Breach** is about one individual live sensor reading, on the **Process Monitoring page**. They never share a score.

KPI Prediction Status — Normal / Warning / Critical

KPI Prediction page

WHAT IT MEANS How far a batch's *predicted* final result is expected to land from the Golden Batch's real outcome, while the batch is still in progress.

FORMULA

$$\text{Deviation \%} = (\text{Predicted} - \text{Golden}) \div \text{Golden} \times 100$$

(direction flipped for Energy/SEC, where lower is better)

15% or more off → Critical
5% or more off → Warning
otherwise → Normal

VARIABLES

Predicted The model's current forecast for the KPI (Yield, Quality, Energy, SEC, or OEE)

Golden The Golden Batch's real recorded result for that same KPI

EXAMPLE

A running batch's Yield is predicted at 93%, versus the Golden Batch's real 99.8% → 6.8% worse than golden → flagged **Warning** (between 5–15%).

Time-to-Breach Process Monitoring page

WHAT IT MEANS An early-warning estimate of how many minutes remain before a drifting reading crosses its limit.

FORMULA

$$\text{Minutes to Breach} = (\text{Limit} - \text{Current Reading}) \div \text{Trend}$$

VARIABLES

Limit The band edge the reading is heading toward

Current Reading The latest live value

Trend How fast the reading is changing per minute, measured over its last 30 minutes

REAL BATCH EXAMPLE

Batch PAR-084's Temperature at minute 200 was 64.7°C, upper limit 64.8°C, trending up by roughly 0.05°C per minute.

Minutes to Breach = $(64.8 - 64.7) \div 0.05 \approx 2$ minutes

Prediction Confidence

[KPI Prediction page](#)

WHAT IT MEANS How much to trust a forecast — combines how much the model's own estimates agree with how far into the batch it is.

FORMULA

Confidence = the LOWER of:

Model Agreement High / Medium / Low, based on how tightly the model's internal estimates cluster

Batch Progress Low – under 25% of the batch has elapsed
Medium – under 55% elapsed
High – 55% or more elapsed

EXAMPLE

A prediction taken only 20% into a batch can never show High confidence — even if the model agrees with itself — because too little of the batch has actually happened yet to be sure.

07 Root-Cause & Diagnosis Confidence

When something deviates, these formulas match it against real historical faults and score how confident that diagnosis is. This is the **Process Monitoring page's** per-parameter "Agent Assessment" panel — a different, separate mechanism from the KPI Prediction page's own contributing-parameters breakdown in Section 06.

Root-Cause Match Score Process Monitoring page

WHAT IT MEANS How well a live deviation pattern matches a known historical fault, using real past batches as evidence.

FORMULA

$$\text{Score} = (\text{Matched Parameters} \div \text{Expected Parameters}) \times \text{Historical Batch Count}$$

VARIABLES

Matched Parameters	How many parameters are trending the direction this fault expects
Expected Parameters	How many parameters this fault pattern involves in total
Historical Batch Count	Number of real past batches that showed this exact fault

EXAMPLE

A fault pattern expects 2 parameters trending a certain way. Both match, and 6 real past batches showed this exact pattern.

$$\text{Score} = (2 \div 2) \times 6 = \mathbf{6.0}$$

It's used internally to pick the best-matching fault before Confidence % (below) is calculated.

Root-Cause Confidence % Process Monitoring page

WHAT IT MEANS How confident the system is in a diagnosis, from how completely it matched and how much real history backs it up.

FORMULA

$$\begin{aligned} \text{Evidence Strength} &= \min(\text{Historical Batch Count} \div 10, 1) \\ \text{Confidence \%} &= 100 \times (0.7 \times \text{Match Fraction} + 0.3 \times \text{Evidence Strength}) \end{aligned}$$

70% or more → High
40% or more → Medium
otherwise → Low

VARIABLES

Match Fraction	Share of expected parameters that matched (0 to 1) — the same Matched Parameters ÷ Expected Parameters ratio from the Match Score above
Evidence Strength	How much historical precedent exists. The real batch count can be any number, less or more than 10 — but its effect here always caps once 10 or more matching batches are on record

EXAMPLE

Full match (Match Fraction = 1.0), backed by 6 historical batches → Evidence Strength = 6 ÷ 10 = 0.6

Confidence = 100 × (0.7 × 1.0 + 0.3 × 0.6) = **88%** → **High**

Shown on Process Monitoring exactly as calculated — e.g. "High (88%)" next to a deviating parameter's assessment.

RECOMMENDED ACTION

The confidence attached to the recommended corrective action is the same number as the root-cause confidence above — the action follows directly from the diagnosis, so it isn't scored separately.

08 Parameters & Units Glossary

The 12 process parameters tracked live, for reference.

PARAMETER	UNIT
Temperature	°C
Process Pressure	bar
Flow Rate	L/min
Agitator RPM	RPM
Assay	%
Inlet Air Humidity	%RH
Exhaust Air Temperature	°C
Filter Differential Pressure	mbar
Shaker Vibration Frequency	Hz
Product Bed Temperature	°C
Chamber Differential Pressure	mbar

PARAMETER	UNIT
AHU Damper Position	%
Compressed Air Pressure	bar